

Fig. 2: Actual-size PCB layout for the monitor circuit

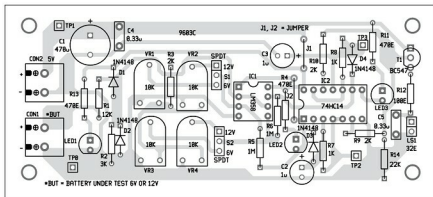


Fig. 3: Component layout for the PCB

amplified by gate N3 and transistor T1 and reproduced by the headphone loudspeaker (LS1) or an earphone. The frequency of the RC oscillator built around gate N2 can be set with the help of resistor R7 and capacitor C2 to an appropriate value.

When the input voltage is below the threshold set by VR3 or VR4, the output at pin OUT2 of IC1 becomes high. The output of gate N4 becomes low and the LED3 indicating a low voltage switches on. The RC oscillator built around gate N5 starts working. The produced square wave signal with frequency of around 1kHz is amplified by gate N6 and transistor T1 and reproduced by the headphone loudspeaker (LS1) or an earphone. The frequency of the RC oscillator built around gate N5 can be set with resistor R8 and capacitor C3 to an appropriate value.

When the input signal is within the range set by the selected potentiometers, both the oscillators do not work, and there is no sound from the

loudspeaker and the LEDs LED2 and LED3 remain off.

Power supply for the circuit may be derived from 6V/12V batteries (not shown) with 5V and ± 2 per cent tolerance and connected at CON2. LED1 is the power indicator.

Construction and testing

An actual-size, single-side PCB for the voltage monitor for 6V/12V batteries is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on a PCB, enclose it in a suitable plastic box.

Fix CON1 on the front side of the case for connecting the battery under test (BUT). Fix CON2 on the rear side of the box for connecting a 5V power supply to the circuit.

Before use, check that voltages at the test points are as per the table. ●

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